

CBL Challenge Based Learning

Trailblazer

Proposed topics: Git branching workflows (each partner creates certain parts of code that are then merged into the final project), Game Design (make the levels challenging enough but understandable to a new player)

Description

Trailblazer is a turn-based puzzle game in which the player controls a small robot trying to escape grid-based rooms. Every tile the player leaves becomes an arrow pointing in the direction of movement. The player must create a trail that guides the robot onto a pressure plate. Each robot has a color-coded pressure plate that when activated, sends the robot on its journey. The robots and player can activate any pressure plate and once all are activated, the door to exit the level unlocks. The first robot should be set in motion by the player which then leads to a chain reaction of robots moving to unlock the door. Once a robot is on a pressure plate, the robot is locked in place. The player can also override the arrows by walking over a tile again.

Product Backlog (in order of priority)

Name: Display a level

Display the room and its game objects in a grid.

How to demo:

Launch the application. Check that the grid shows walls, player, robots, pressure plates and locked exit. The initial robot direction matters as it's the first move they will make.

Name: Move the player

Allow the player to move one tile at a time using the arrow keys.

How to demo:

Press each arrow key. Check that the player moves one tile. Attempt to move into a wall or outside the grid. Block invalid moves. Invalid moves do not advance the turn.

Name: Create and overwrite arrows

Create a direction arrow whenever the player leaves a tile and replace any existing arrow on that tile.

How to demo:

Move right from a tile: check that it displays a right arrow. Return to that tile and leave upwards: check that the arrow changes to up.

Name: Active Pressure Plates

Initiate movement when a robot's pressure plate is pressed.

How to demo:

Walk over a robot's pressure plate. That robot should initiate movement.

Name: Robot Movement

Robot moves in the direction of the arrow below it. If a robot is on a pressure plate it should stop moving.

How to demo:

Create an arrow path towards a pressure plate and walk over a robot's pressure plate. The robot should start moving along that path. The first robot move should be in the initial direction it is facing. Once the robot reaches the pressure plate it should stop.

Name: Unlock Exit

Unlock the exit once all pressure plates are activated.

How to demo:

Activate all pressure plates on the level and check whether the exit opens.

Name: Restart Button

A button on the side of the grid can always be pressed to restart the level.

How to demo:

Press the restart button and check whether it restarts the level. All entities return to initial position, pressure plates deactivate, door locks and arrows disappear.

Name: Level Passed

When player reaches the unlocked exit, the level is completed.

How to demo:

Complete the level, enter the exit and check whether a victory screen appears.

Name: Progress through levels

On the victory screen there will be a button to advance to the next level.

How to demo:

Complete a level and check whether the button appears on the victory screen. Press the button and check whether a new level loads in.